

Yuli Multi-Enhancing Artificial Vision by Integrating Machine Learning in Retinal Prosthetics

RWTHAACHEN
UNIVERSITY

3

0

Visual Fixation

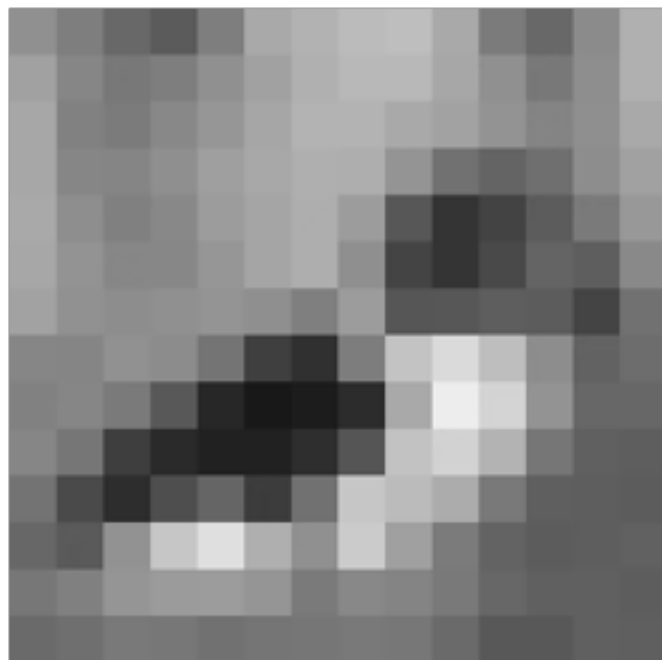
224x224



Input Image



14x14



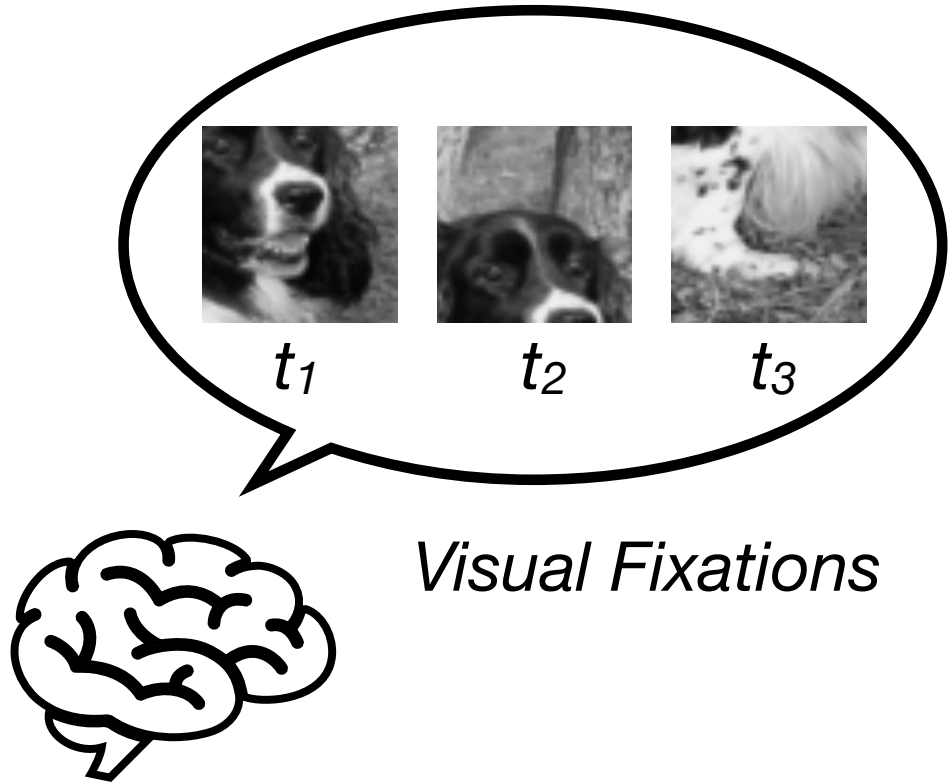
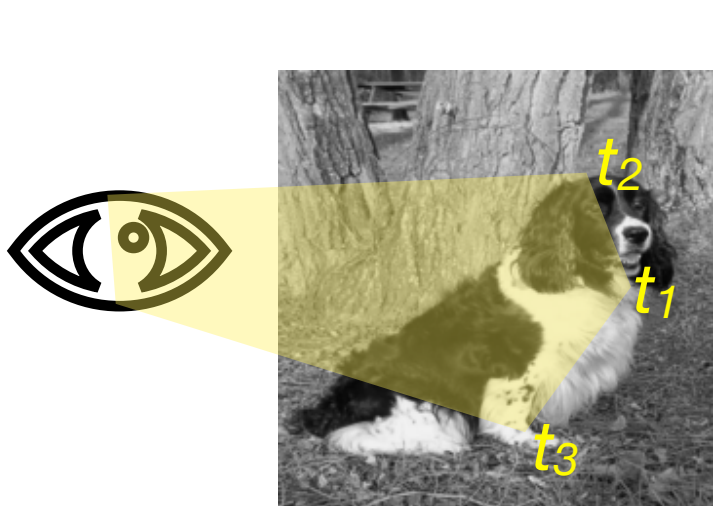
Stimulus



Percept



[Mu et al., 2025a, Mu et al., 2025b]



0.00s



- **Biomimetic:** How we see
- **Practical:** Low resolution
- **Transferable:** Consistency evaluation







<https://www.projectaria.com>

Aria Glasses. Granted by Meta Inc.

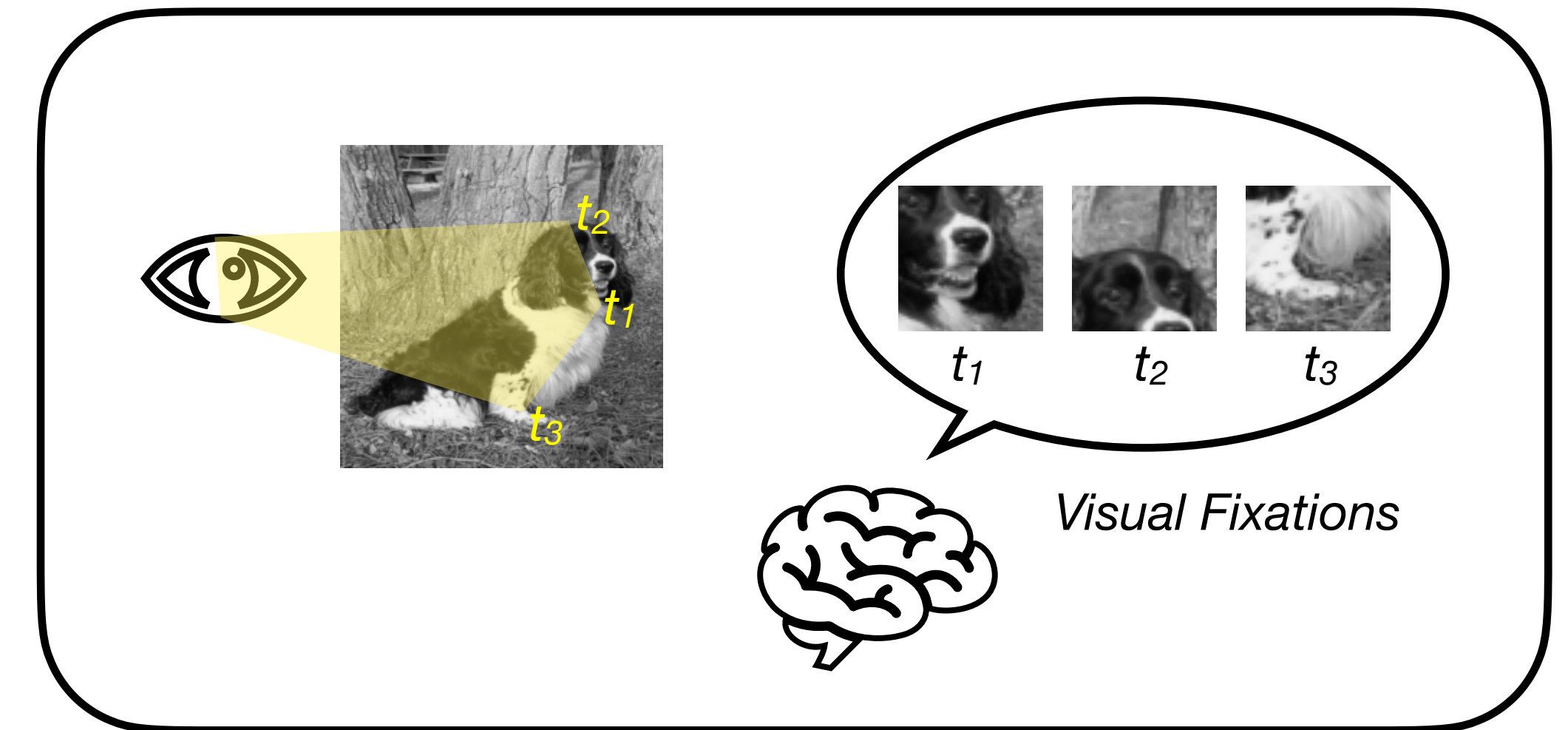
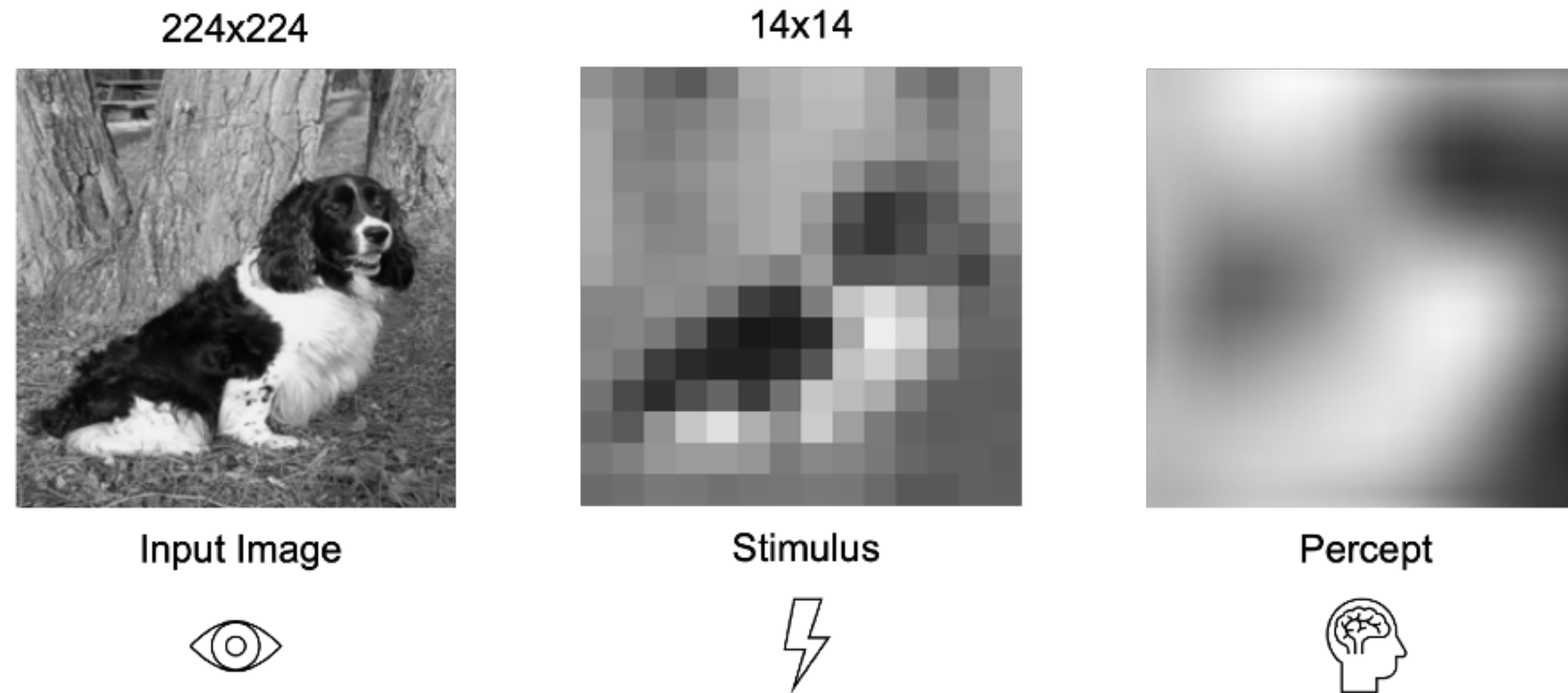
0.00s



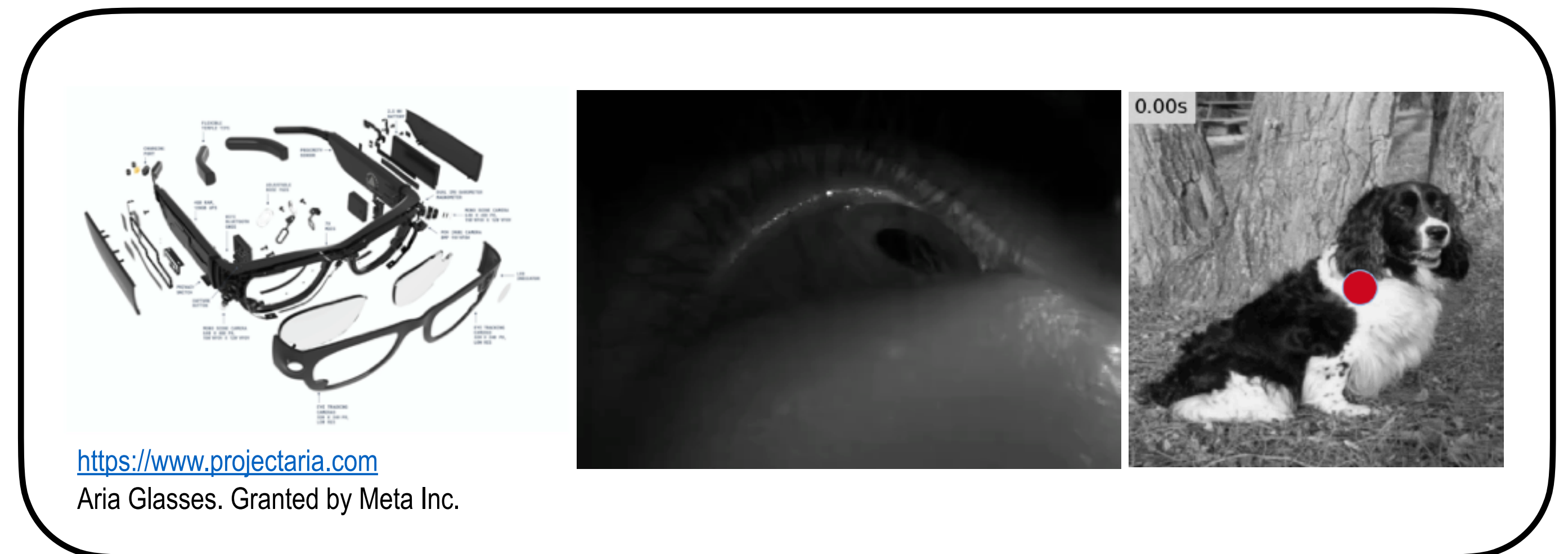
0.00s



Visual Fixation

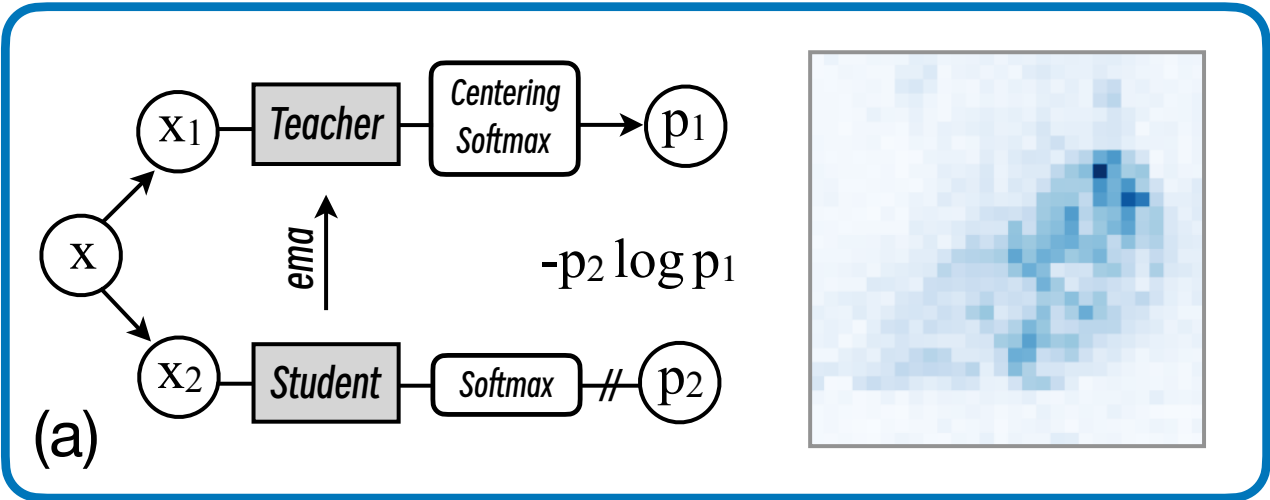
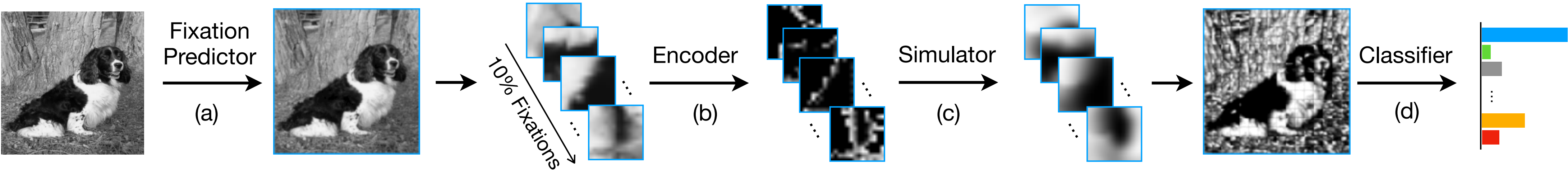


- **Biomimetic:** How we see
- **Practical:** Low resolution
- **Transferable:** Consistency evaluation

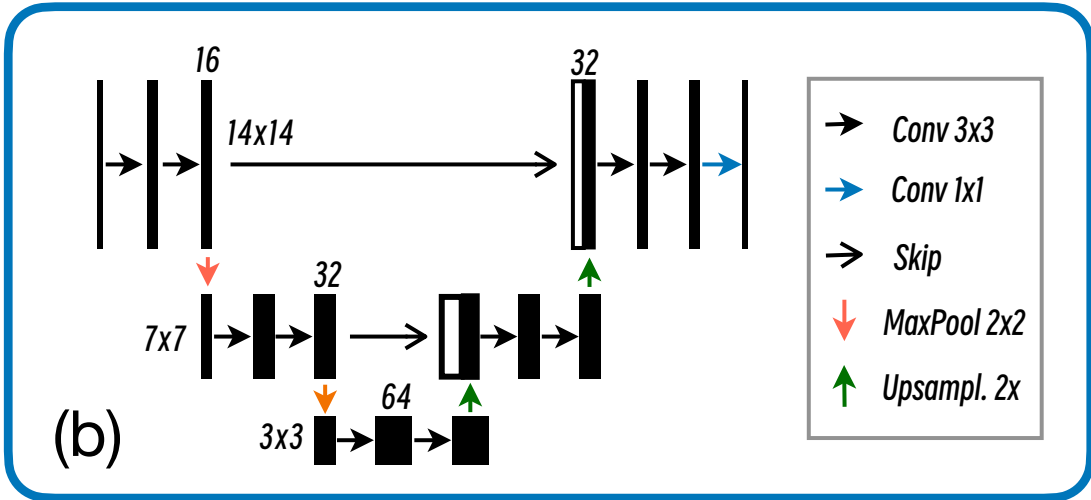


[Wu et al., 2025a, Wu et al., 2025b]

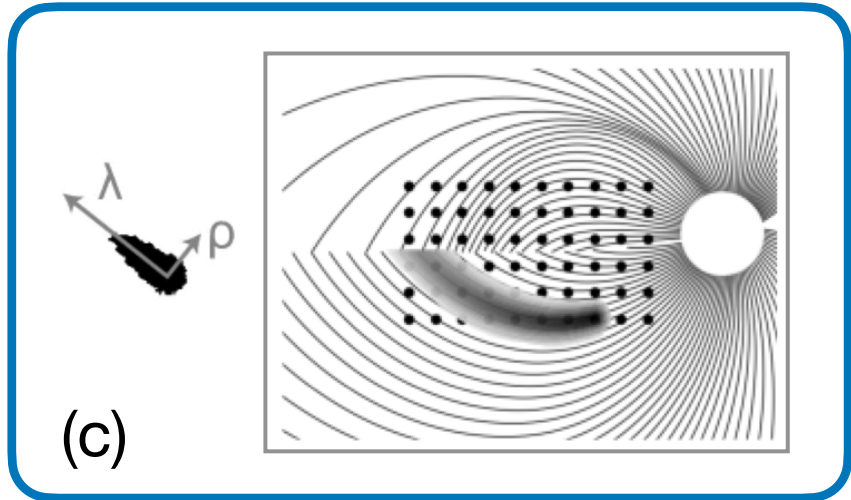
Visual Fixation-based Simulation



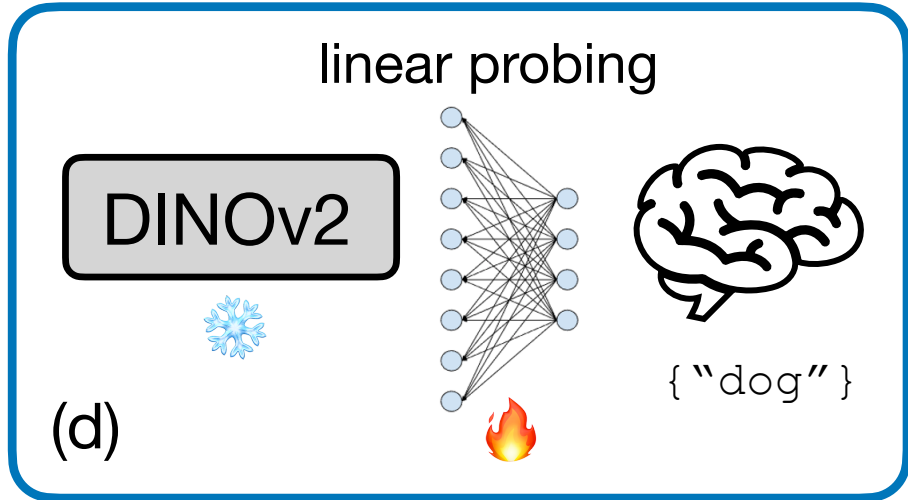
Fixation Predictor: Attention map of ViT



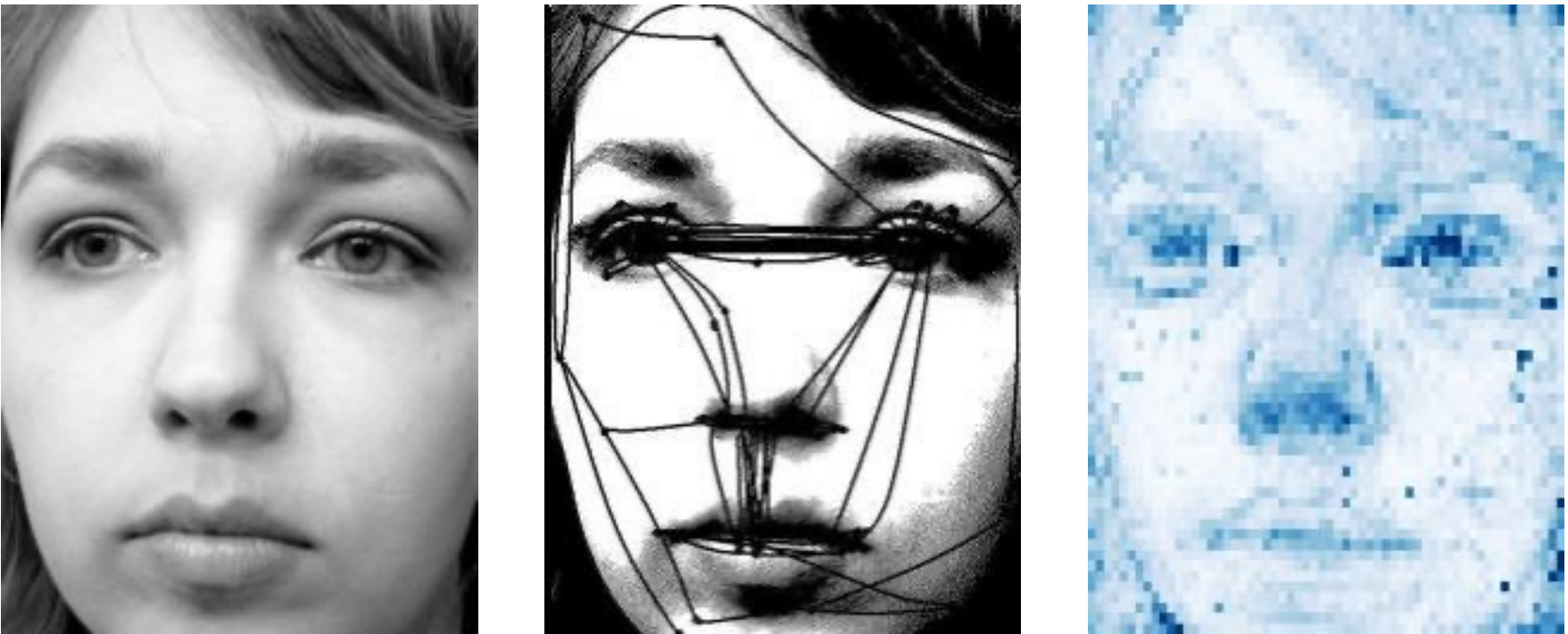
Stimulus Encoder: U-Net



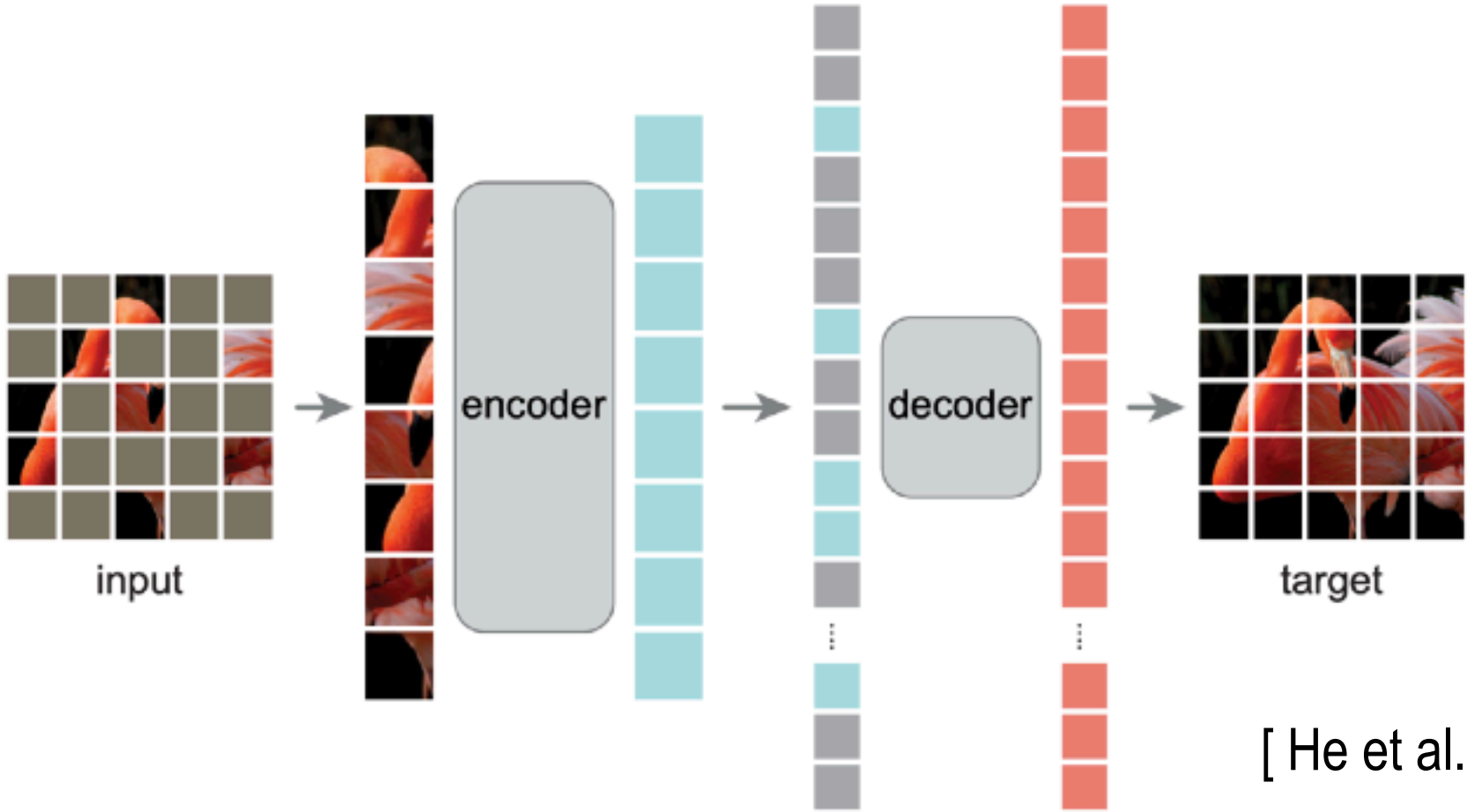
Percept Simulator: pulse2percept



Classifier: DINOv2



CC BY-SA 2.0, via Wikimedia Commons



[He et al., 2021]

[Wu et al., 2025a]